

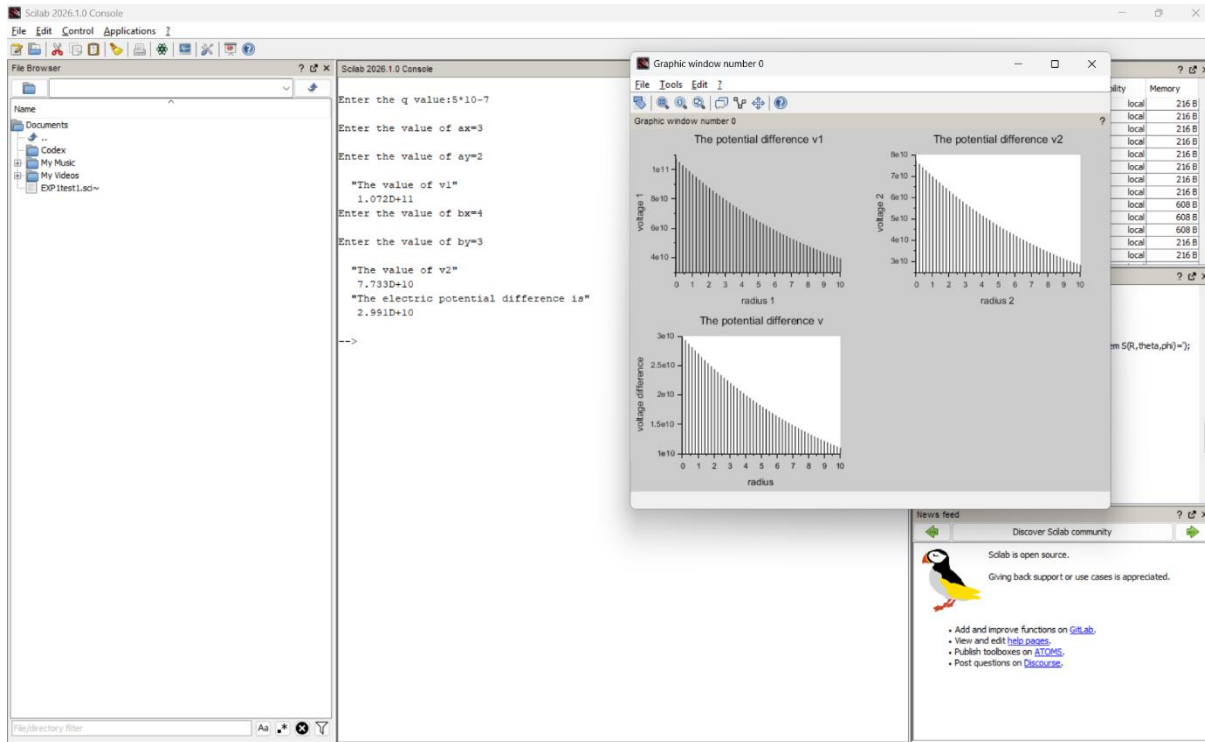
EXP 5: Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

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1 clear;
2 clear;
3 q=input('Enter the q value:');
4 e=8.854*10^-12;
5 pi=3.14;
6 ax=input('Enter the value of ax=');
7 ay=input('Enter the value of ay=');
8 z1=sqrt(ax^2+ay^2);
9 v1=q/(4*pi*e*z1);
10 disp('The value of v1',v1);
11 bx=input('Enter the value of bx=');
12 by=input('Enter the value of by=');
13 z2=sqrt(bx^2+by^2);
14 v2=q/(4*pi*e*z2);
15 disp('The value of v2',v2);
16 r=r1-z2;
17 v=v1-v2;
18 disp('The electric potential difference is',v);
19 r1=linspace(0,10,50);
20 r2=linspace(0,10,50);
21 r=linspace(0,10,50);
22 y=v1*exp(-0.1.*r1);
23 z=v2*exp(-0.1.*r2);
24 h=v*exp(-0.1.*r);
25 figure();
26 subplot(3,2,1);
27 plot(r1,y);
28 xlabel('radius-1');
29 ylabel('voltage-1');
30 title('The potential difference v1');
31 subplot(3,2,2);
32 plot(r2,z);
33 xlabel('radius-2');
34 ylabel('voltage-2');
35 title('The potential difference v2');
36 subplot(3,2,3);
37 plot(r,h);
38 xlabel('radius');
39 ylabel('voltage difference');
40 title('The potential difference v');

```

Line 40, Column 36.



Exp 5:-

$$q = 5 \times 10^{-10} \text{ C}$$

$$ax = 3$$

$$ay = 2$$

$$bx = 4$$

$$by = 3$$

$$\text{distance} = r_1 = \sqrt{ax^2 + ay^2}$$

$$= \sqrt{3^2 + 2^2}$$

$$= \sqrt{9 + 4}$$

$$= \sqrt{13}$$

$$r_1 = 3.6056 \text{ m}$$

$$V_1 = \frac{q}{4\pi\epsilon_0 r_1} = \frac{43}{4 \times 3.14 \times (8.85 \times 10^{-12}) \times 3.6056}$$
$$= \frac{43}{4.012 \times 10^{-10}}$$

$$V_1 = 1.072 \times 10^{10} \text{ V}$$

$$V_2 =$$

$$r_2 = \sqrt{bx^2 + by^2}$$

$$= \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25}$$

$$= 5 \text{ m}$$

$$V_2 = \frac{43}{4 \times 3.14 \times (8.85 \times 10^{-12}) \times 5}$$
$$= \frac{43}{5.259912 \times 10^{-10}}$$

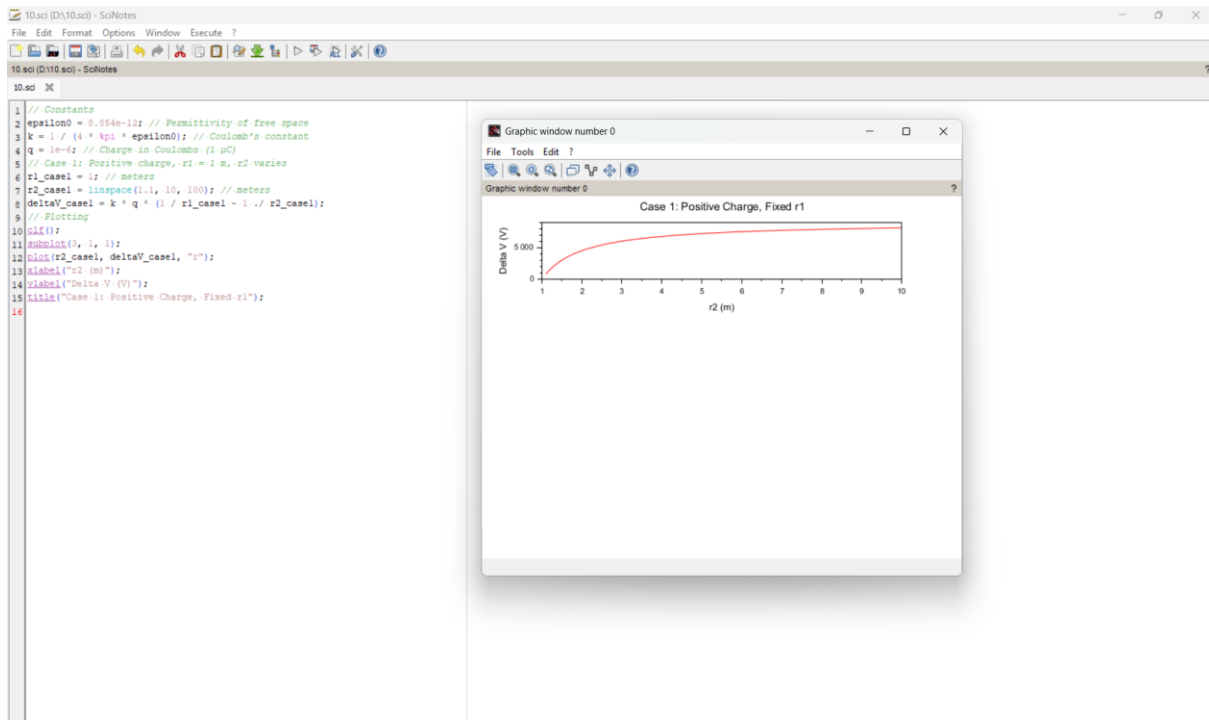
$$V_2 = 7.733 \times 10^{10} \text{ V}$$

Electric potential difference, $V = V_1 - V_2$

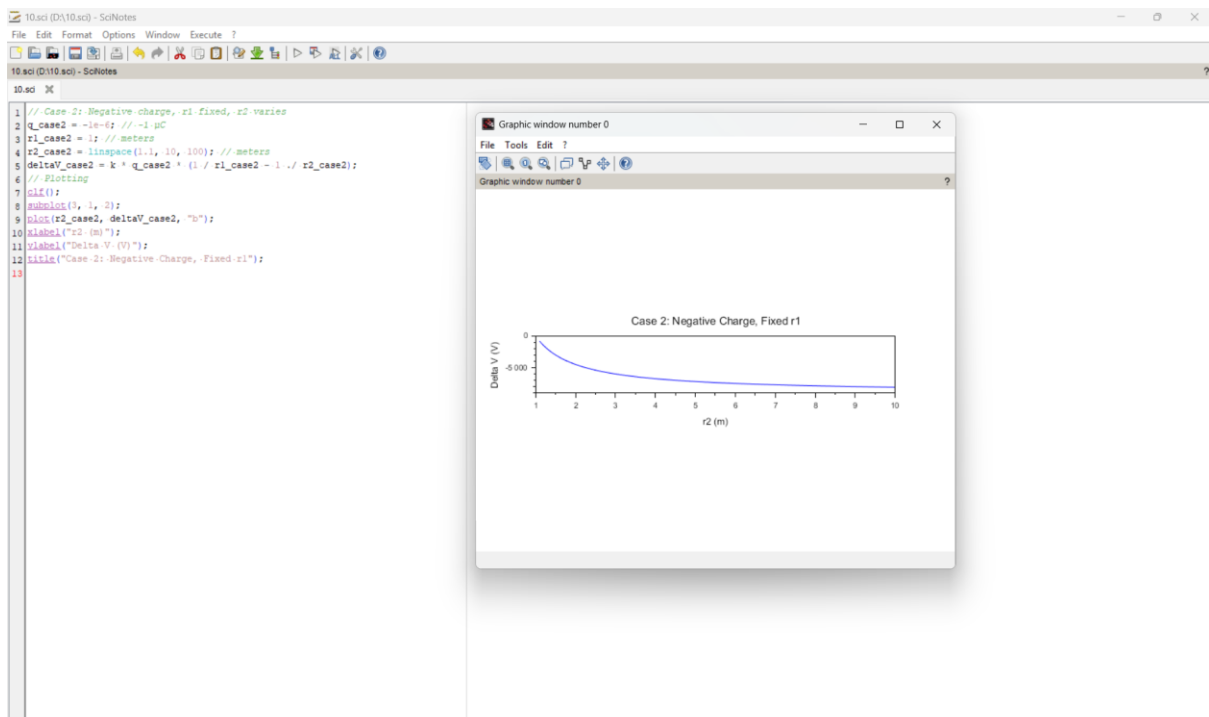
$$(1.072 \times 10^{10}) - (7.733 \times 10^{10})$$

$$V = -2.991 \times 10^{10} \text{ V}$$

Case 1: A positive charge with fixed $r_1=1$ m r_2 varying.



Case 2: A negative charge with r_1 fixed and r_2 varying.



Case 3: Both points r_1 and r_2 vary with a positive charge.

